

Business Concept Generation

Group 9

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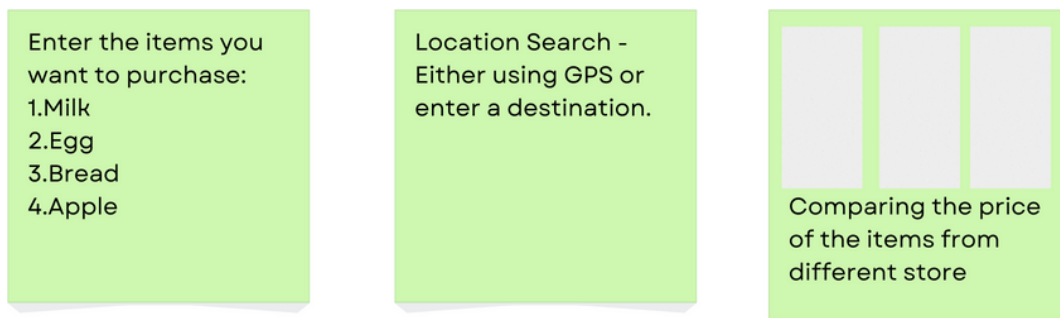
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Customer Pain 1: Customers spend too much time finding affordable food and only want to visit some stores in person to afford the food.

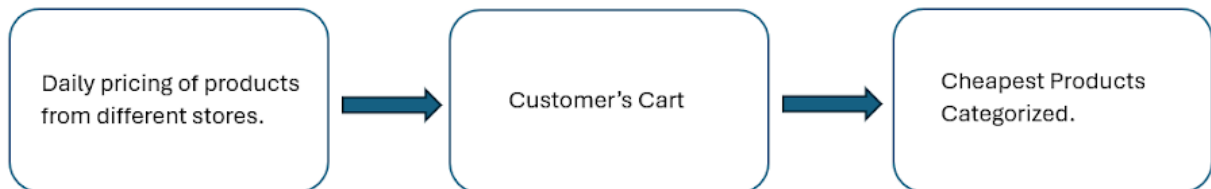
Problem Definition 1: Customers are spending too much time searching for cheap food and only want to go to specific stores in person to find affordable options. This means there's a problem with finding budget-friendly dining choices efficiently. Customers are looking for a simpler way to decide where to get affordable food.

Solution 1:



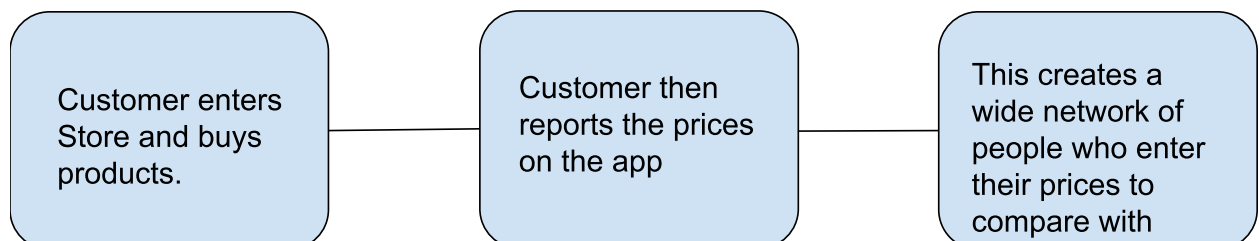
An application can be used as an effective tool for this. The application works by comparing the prices from different stores. The customer can input the things they wish to buy and the app compares the price items from all the stores and gives them the low-price option from all stores.

Solution 2:



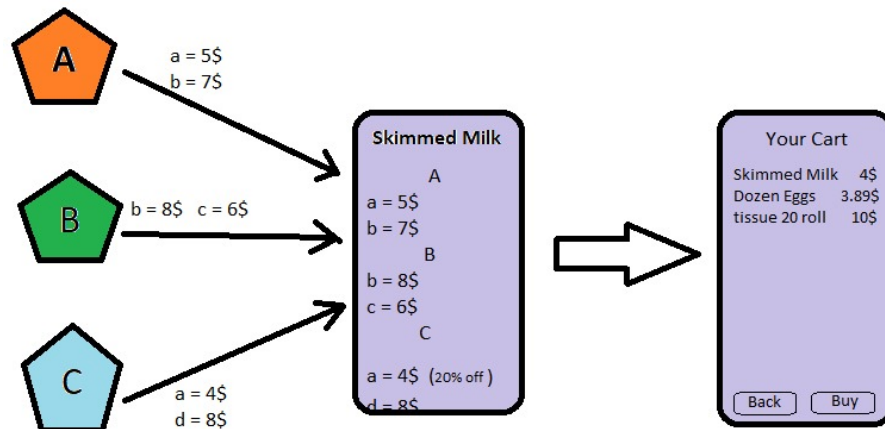
A user-friendly application designed to store and compare product prices from nearby stores conveniently. As customers add items to their cart, the app categorizes and displays the pricing information from various stores in its database, along with the availability of the products in each store.

Solution 3:



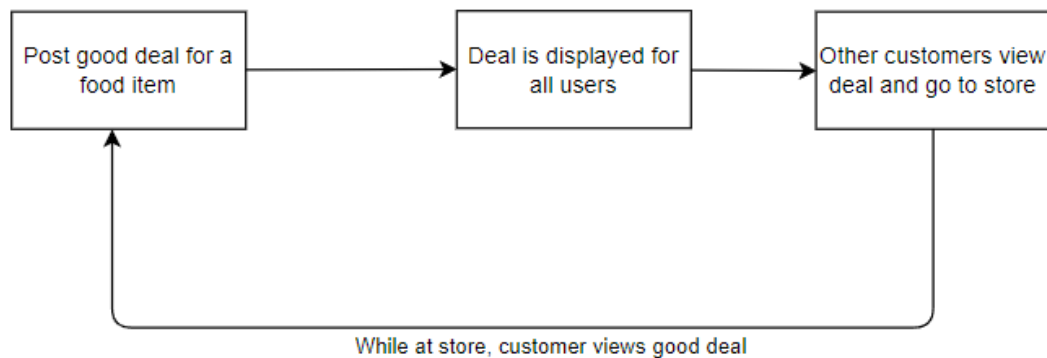
A user based application that allows indirect communication between the clientele to allow price comparison for future reference. Through this app people can see the various prices without having to enter the store and to make their decision on which store to go to.

Solution 4:



An app which takes real time data from stores within the specified distance from the user location. The app will show the prices of the specified product (Eg. A dozen Egg) of different brands from each store. The user will select the item that he/she is looking for. After getting all the items in the cart, he/she can either buy it in person or buy it online.

Solution 5:

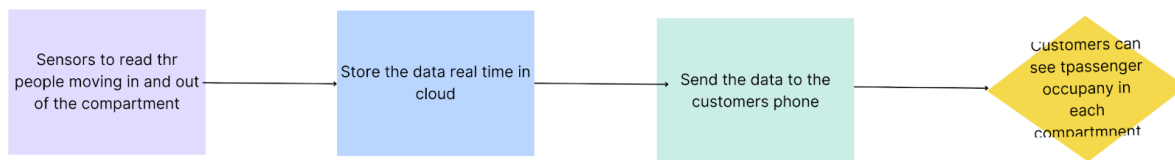


An application, which uses user-defined data to populate its content, that shows the cheapest prices for different kinds of foods. It will rely on users to provide “good deals” and then this will be listed and displayed for others to see. This is similar to what has been done for a similar app designed to find the cheapest gas in your local area.

Customer Pain 2: Travelers spend too much time standing on the train without knowing the spaces available on the other sections of the train.

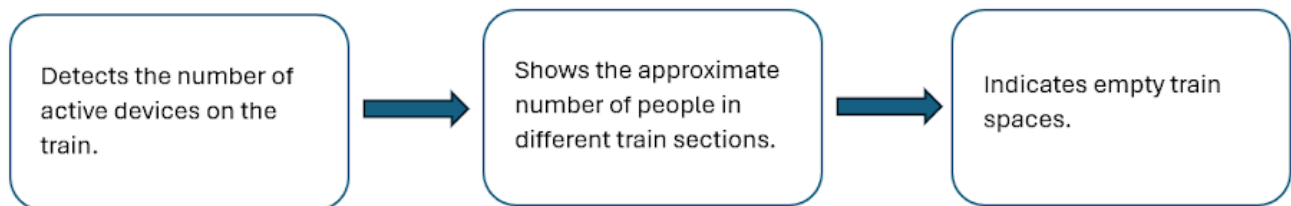
Problem definition 2: People who travel on trains spend a lot of time standing because they don't know if there's space in other parts of the train. This makes the journey uncomfortable and not easy for travellers. People want a solution that can help them know where there is space on the train, so they can have a more comfortable journey without standing for too long.

Solution 1:



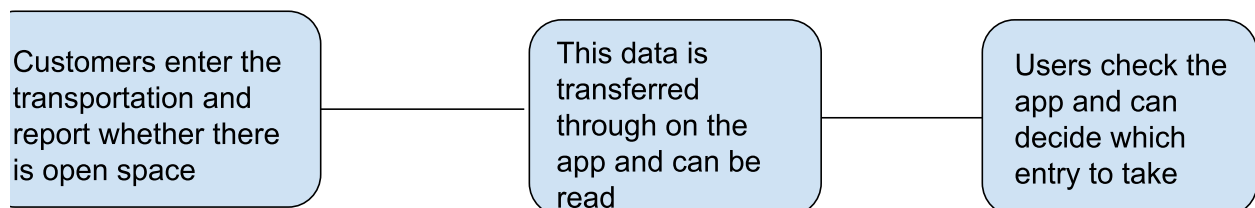
The solution would be checking the person's move out and in a compartment. In doing so the number of people present in a compartment can be found out. This information can be sent via cloud to individual users via application. Thereby the customers can foresee which compartment has less congestion and travel more comfortably.

Solution 2:



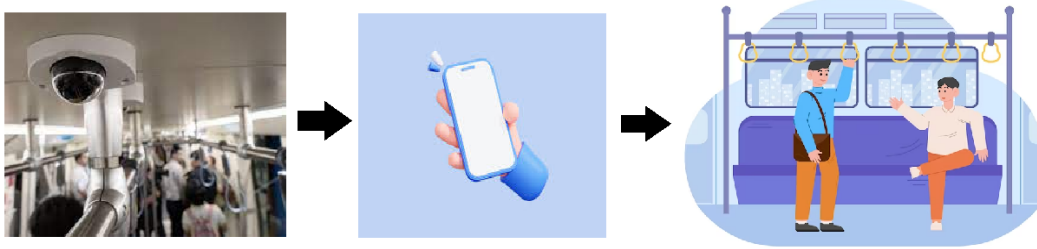
An application designed to determine the number of active devices on a train, providing an estimate of the crowd density in a particular location. This app also identifies vacant spaces within the train, guiding users to the most optimal door for entry to minimize exposure to crowds.

Solution 3:



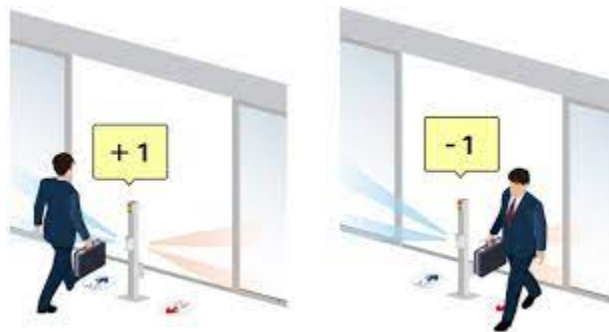
This would be an application which would allow people to quickly assess where to sit in public transportation. Through the network, people can quickly report data which will help people make their decision.

Solution 4:



Each train compartment has a CCTV camera. It will use an algorithm to count the number of people in the compartment. The app will collect data gathered from the CCTV camera and make a miniature train which shows the number of passengers in each compartment and the passenger using this app can get to know which compartment is less crowded.

Solution 5:

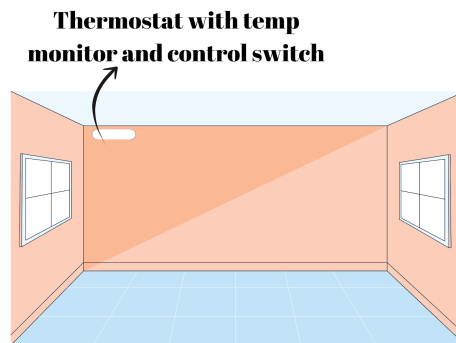


A possible solution would be to develop an application that gives users information on how busy different train cars are, and even the platform. This would be accomplished by placing sensors in the train and on the platform. The customer would be the cities and the companies who manage the trains in the cities.

Customer Pain 3: Customers often find it difficult to modify room-by-room temperature in their apartment.

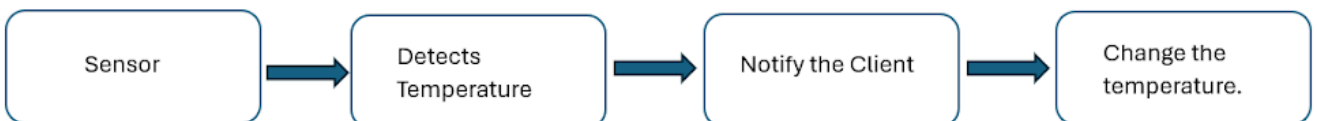
Problem definition 3: Customers have trouble adjusting the temperature in different rooms of their apartments. This makes it uncomfortable because some rooms might be too hot or too cold. People want an easier way to control the temperature in each room so they can be more comfortable in their own homes. Customers want to be able to easily change the temperature in different rooms.

Solution 1:



Individual room temperature control is possible with zoned heating and cooling. This strategy involves partitioning your house into zones and putting a separate thermostat in each one. Each thermostat may then manage the temperature in its own zone, allowing you to personalize the settings for each area.

Solution 2:



Each room is equipped with a sensor integrated with the heater, that detects and monitors the temperature. This system notifies the customer through an app, allowing them to customize the temperature settings for individual rooms according to their preferences, ensuring room temperature is regulated.

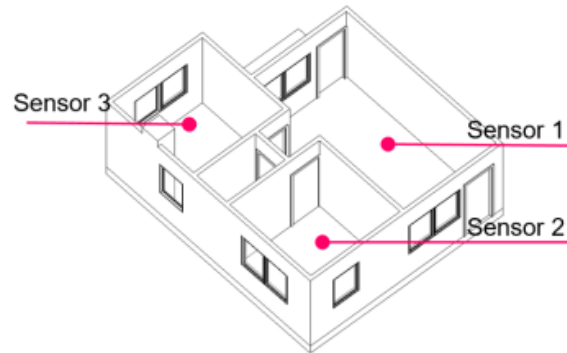
Solution 3:



This would place individual sensors on each system which could detect the various temperatures and adjust the heating. This adjusted heating would be based on the default

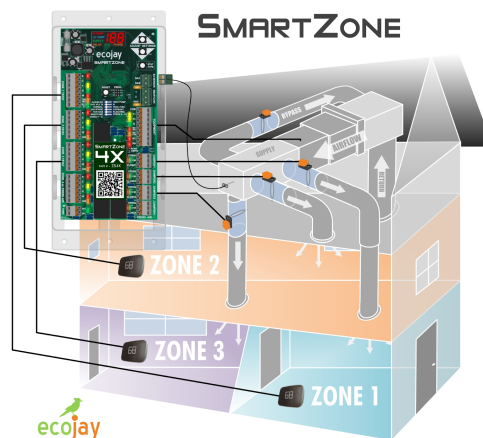
settings set on the app by the customer. The customer could also with the help of the app change the conditions inside of the house or apartment.

Solution 4:



Placing a temperature sensor (Eg. thermocouple) in each of the rooms that gauges the temperature of the room every 10 mins. When the temperature of the house is set to a certain degree (Eg. 72°F/22°C) the sensor will retrieve the temperature data of each room and then the main heating system will use this data to adjust the temperature of the heaters in all the rooms to maintain a constant temperature around the house.

Solution 5:



A solution to this is to place sensors in different rooms and this will create different zones in the house. Each zone will be heated/cooled independently from other zones. This will allow for the users to have full control over the temperature in each room, whereas before it rooms would be different temperature based on the size and air flow present.

Concept Evaluation

Problem 1: Customers spend too much time finding affordable food and only want to visit some stores in person to afford the food.

In addressing the challenge of finding affordable food, Solution 1, a price comparison application, hinges on user engagement, requiring a critical mass for effectiveness. Solution 2, a user-friendly app for comparing prices, has potential for broad user adoption. Solution 3, a user-based app for indirect communication and price comparison, relies on user participation. Many apps have tried to give customers certain features like finding nearby offers, or maybe comparing different products in the same store. So a combination of the solutions and already existing features and developing these in one app is an opportunity. The external context of increasing demand for budget-friendly dining favors these solutions, but their success depends on creating a user-friendly experience. Funding can come from venture capitalists, with deal terms requiring a balance between control and attracting quality investors. Risks include dependence on user-generated content and shifts in user behavior or market dynamics.

References: <https://www.plotprojects.com/blog/local-deals-apps/>

Problem 2: Travelers spend too much time standing on the train without knowing the spaces available on the other sections of the train.

Solutions 1 and 2 address train crowding through crowd-sourced data and device tracking, requiring widespread user adoption. Solution 4, utilizing CCTV cameras, faces privacy concerns. There are already existing projects which aim at predicting real time train crowdedness through different means, like the app NS. Although such projects in Canada are far fewer. This is an opportunity. The external context, marked by urbanization and crowded public transport, favors these solutions. Funding sources could include public transportation authorities or tech investors. Founders must balance deal terms with financial support and user experience integrity. Risks involve privacy concerns, user adoption, and reluctance from transportation authorities to invest in new technologies.

References:

<https://towardsdatascience.com/real-time-crowdedness-predictions-for-train-travelers-a5a5d2858bed>

<https://www.wired.co.uk/article/train-crowds-app>

<https://venturebeat.com/mobile/google-maps-now-predicts-how-crowded-your-bus-or-train-will-be/>

Problem 3: Customers often find it difficult to modify room-by-room temperature in their apartment.

Solutions 1 and 2, enabling individual room temperature control, depend on user interaction. Solution 4, using sensors and data, requires user trust in technology. The external context, with a growing demand for smart home solutions, aligns well with these temperature control innovations. As there are already existing products, but more expensive, a possible opportunity would be making it cheaper and available for more people. Funding could come from smart home device manufacturers or IoT investors. Deal terms should balance financial support with strategic partnerships. Risks include user acceptance of IoT devices, technical glitches, data privacy concerns, and a slow adoption rate. Careful consideration of these factors is crucial for success.

References:

<https://www.warmup.ca/blog/best-smart-thermostats-for-multiple-zones#:~:text=Ecobee,adjusts%20to%20keep%20everyone%20comfortable>